BPG-13 · BEST PRACTICE GUIDES

Cash flow forecasting

The gap between doing the work and having the money

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Summary

A cost S-curve says nothing about whether a project can pay its people next month. This guide sets out the four mechanisms that separate progress from cash — measurement and certification, payment terms, retention and advance recovery — builds a monthly forecast through the certificate cascade, identifies the peak funding requirement, shows why a profitable project can still fail on cash, and runs the payment-timing sensitivity that turns a forecast into a decision about facilities.

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Contents

1. The S-curve is a cost curve ... 4
2. The four gaps ... 4
3. Building the forecast: the certificate cascade ... 5
4. The outflow side ... 5
5. The funding trough ... 6
6. Why a profitable project fails on cash ... 6
7. Sensitivity: change one thing at a time ... 7
8. Timing effects that are not costs ... 7
9. How this goes wrong ... 7
10. Worked example ... 8
10.1 The certificate cascade ... 8
10.2 The cash position, month by month ... 9
10.3 Profitable, and needing 2.08 million ... 9
10.4 Sensitivity 1 — payment one month later ... 9
10.5 Sensitivity 2 — no mobilisation advance ... 10
11. Checklist ... 11
Related ... 12
Sources and standards ... 12
Status and version ... 12

Cash flow forecasting

In one paragraph. A cost S-curve says nothing about whether a project can pay its people next month. This guide sets out the four mechanisms that separate progress from cash — measurement and certification, payment terms, retention and advance recovery — builds a monthly forecast through the certificate cascade, identifies the peak funding requirement, shows why a profitable project can still fail on cash, and runs the payment-timing sensitivity that turns a forecast into a decision about facilities.

Who this is for. Cost engineers, commercial managers and project controls managers who produce or rely on a project cash forecast, and the finance and treasury colleagues who have to fund the answer.

1. The S-curve is a cost curve

The cumulative cost curve every project draws describes when value is earned and cost is incurred. Cash is a different curve with a different shape, and on most contracts it sits well below and well behind the cost curve for most of the project's life.

The distinction is not academic. A project can be exactly on budget, exactly on programme, and unable to pay its subcontractors, because being owed money and having money are different states. Every mechanism that separates them is contractual, predictable and modellable — which means a funding shortfall is almost always a forecasting failure rather than a surprise.

Two curves are therefore needed, and they answer different questions:

Curve	Question it answers	Driven by
Cumulative cost / earned value	Are we going to be within budget?	Progress and productivity
Cumulative net cash	Can we pay for the work while we do it?	Certification, terms, retention, advances

2. The four gaps

Gap 1 — measurement and certification. Work performed is not work valued. The contractor applies for what it believes the work is worth; the certifier assesses; the certified figure, not the applied figure, is what becomes payable. Persistent under-certification is a commercial issue, but for the forecast it is a simple discipline: **forecast on expected certified value, and hold the application-to-certification gap as a stated assumption** rather than assuming full certification and treating shortfalls as bad luck.

Gap 2 — payment terms. Certified sums are paid after a contractual period running from the certificate or the application. On a monthly cycle the practical lag between performing work and receiving cash for it is usually two months, sometimes three, and the difference between those two is enormous — §9 quantifies it.

Gap 3 — retention. A percentage of each certified amount is withheld as security, released in stages, often long after the work is done and partly after a defects period. Retention converts margin into a deferred receivable. A project can finish, hand over, be paid everything payable, and still be cash-negative until release.

Gap 4 — advance recovery. Where a mobilisation advance is paid, it is a loan repaid through the measure — recovered as a deduction on each certificate, usually at a stated percentage of the gross value. The forecast must carry both the early inflow and the reduced net certificates that repay it, or the advance looks like free money in the first month and an unexplained shortfall thereafter.

Together these four produce the shape every project cash curve has: a trough, whose depth and timing are the things that actually need managing.

3. Building the forecast: the certificate cascade

The inflow side of the forecast is one repeated calculation, per period:

```
Gross value certified        (from the valuation of work done)
 − retention                 (stated % of gross)
 − advance recovery          (stated % of gross, until the advance is repaid)
 − previous payments         (where the valuation is cumulative)
 = net amount due
 → received on the payment date given by the contract terms
```

Three modelling rules make it reliable:

Model the value, then the deductions, then the date — separately. Collapsing them into one net figure per month makes the forecast impossible to interrogate when it goes wrong, and it will go wrong.

Phase from the schedule, not from the calendar. Gross value per period comes from the time-phased baseline and the current forecast of progress. A cash forecast built by spreading the contract value evenly across the programme is not a forecast.

Carry retention forward explicitly, with its release dates. Retention is not a cost and it is not lost; it is cash held. It belongs on the forecast as a receivable with dates, and somebody must be accountable for recovering it. Retention that nobody chases is a real and recurring loss.

4. The outflow side

Outflows are governed by different terms from inflows, and the mismatch is the whole problem.

- **Labour** is paid weekly or monthly, in arrears of days rather than months. It is close to uncompressible.
- **Subcontractors** are paid on their own terms. Where those terms are shorter than the terms on which the main contract pays, every period of work is being funded by the contractor.

- **Materials and equipment** are paid on supplier terms, sometimes with deposits or stage payments before delivery — an outflow that precedes any possibility of certifying the corresponding value.
- **Plant and preliminaries** run at a rate per period regardless of progress, which is why they dominate the cash consequence of any delay.

The two levers on this side are supplier payment terms and the timing of procurement commitments. Both have limits: stretching supplier terms transfers a funding problem down the supply chain, has commercial consequences, and in some jurisdictions is regulated. Treat it as a lever with a cost, not as free financing.

5. The funding trough

The deepest point of the cumulative net cash curve is the **peak funding requirement** — the amount of money the project must have available, from a facility, from group treasury, or from partners, in order to keep operating. It is a single number, it has a date, and it is the most useful output of the whole exercise.

Its drivers are precisely the levers a commercial and controls team manages: certification performance, payment terms both ways, retention percentage and release profile, billing cadence, advance payments and their recovery rate, and the margin itself.

Two consequences follow that reports rarely state:

The trough is a decision, not a fact. Every driver in that list is negotiable at some point — at tender, at award, or in the running of the job. A forecast that presents the trough as a given has skipped the conversation that matters.

The trough moves for reasons that have nothing to do with performance. A perfectly performing project whose client moves from 30-day to 60-day payment has a materially worse funding position and an unchanged cost report. That divergence is exactly why cash is reported separately.

6. Why a profitable project fails on cash

Profit is measured on value earned against cost incurred. Cash is measured on money received against money paid. On a contract of any length the second is far more volatile than the first, for a structural reason: cost is incurred continuously and revenue is collected in lumps, late, net of deductions.

Three patterns account for most failures.

Growth. A contractor winning more work funds each new project's trough before the previous one has recovered. The business is profitable and getting steadily closer to insolvency. This is why the aggregate funding profile across a portfolio matters as much as any single project's.

Front-loaded cost, back-loaded value. Design, mobilisation, temporary works, procurement deposits and long-lead equipment are paid early and certified late — or, in some structures, not certified as value at all until incorporated into the works.

Retention plus a long defects period. The entire margin can be smaller than the retention held. Where that is true, the project's profitability is contingent on a release that happens a year after everyone has moved on.

7. Sensitivity: change one thing at a time

A cash forecast that is presented as a single line is not decision-ready. Run a small, disciplined set of sensitivities, each moving exactly one variable, and report the effect on two numbers only: the depth of the trough and its date.

The four worth running on almost every project:

1. **Payment terms slip by one period.** The single most common and most damaging change.
2. **Certification at less than application.** A stated percentage, with the balance certified later.
3. **No advance, or a slower advance.** Tests how much of the funding position depends on it.
4. **Retention released late.** Moves the end of the curve rather than the trough, but decides when the margin becomes real.

Moving one variable at a time is not a stylistic preference. A forecast that changes three assumptions simultaneously produces a number nobody can attribute, and attribution is the entire purpose — the point is to know which lever to pull.

8. Timing effects that are not costs

Several flows move cash without changing project cost, and they belong in the forecast for their timing alone.

Where an indirect tax such as a value-added or goods-and-services tax applies, invoices are issued and paid gross of it, and the tax then leaves again on its own remittance calendar. Whether such a tax applies, at what rate, on what basis and with what remittance timing depends entirely on the jurisdiction and is a question for the finance function — but where it applies, the gross flows can be material at the trough. Similarly, where withholding applies to cross-border payments, the total cash is unchanged but its counterparties and dates are not. Neither affects project cost; both affect the funding requirement.

The discipline is to model gross flows with their remittance dates, and never to let a tax-inclusive invoice value reach the cost ledger.

9. How this goes wrong

The cash forecast is the cost curve with a lag applied. A single blanket delay applied to the S-curve, with no retention, no advance recovery and no certification assumption. It will be wrong by the size of the deductions, which is the size of the problem.

Forecasting on applications rather than certificates. Optimistic by exactly the assessment gap, every month, cumulatively.

Retention modelled as lost or forgotten entirely. Either the forecast understates the eventual position or it overstates the current one. Model it as held cash with release dates and an owner.

The advance treated as income. It arrives, the curve lifts, and nobody models the recovery deductions that follow. The second and third months then look inexplicably poor.

Ignoring the outflow terms. Modelling receipts carefully and assuming all costs are paid in the month incurred understates the position on projects with long supplier terms and overstates it where subcontractors are paid quickly.

No sensitivity on payment timing. The one assumption most likely to change, and the one most often presented as fixed.

Unpriced variations invisible in the cash line. Work instructed but not agreed is being funded with nothing certified against it. A growing unpriced book is a cash problem before it is a commercial one — see `BPG-11 — Change orders and variations`.

One project at a time. The funding requirement that matters to a business is the aggregate of its projects' troughs at each point in time, which is not the sum of their peak requirements and is never visible from any single project's report.

10. Worked example

Illustrative figures. Currency USD; monthly periods; a six-month package of contract value 6,000,000; retention 5 % of gross certified value; mobilisation advance 8 % of contract value, paid at the start of Month 0 and recovered at 8 % of each gross certificate; the certifier is assumed to certify the full applied value; payment is received two months after the month in which the work is performed; all outflows are assumed paid in the month the cost is incurred; total cost 5,400,000, giving a margin of 600,000, which is 10 % of contract value. Figures are rounded to the nearest whole unit and are exact as stated.

10.1 The certificate cascade

Month	Gross value certified	Retention 5 %	Advance recovery 8 %	Net amount due
1	500,000	25,000	40,000	435,000
2	900,000	45,000	72,000	783,000
3	1,300,000	65,000	104,000	1,131,000
4	1,500,000	75,000	120,000	1,305,000
5	1,100,000	55,000	88,000	957,000
6	700,000	35,000	56,000	609,000
Total	**6,000,000**	**300,000**	**480,000**	**5,220,000**

Two reconciliations to run every time, because they catch most modelling errors:

```
Retention total      = 5 % × 6,000,000 = 300,000 ✓
Advance recovered    = 8 % × 6,000,000 = 480,000 = the advance paid ✓
Net certified total  = 6,000,000 − 300,000 − 480,000 = 5,220,000 ✓
```

The second one matters: the recovery percentage must be set so that the advance is fully repaid by the end of the works. Set it too low and the final certificates carry a balance nobody has planned for.

10.2 The cash position, month by month

Costs are `0.9 × gross value` in each month, paid in the month incurred. Receipts arrive two months after the work month.

Month	Cash in	Cash out	Net in month	Cumulative cash
0	480,000 (advance)	—	480,000	480,000
1	—	450,000	(450,000)	30,000
2	—	810,000	(810,000)	(780,000)
3	435,000	1,170,000	(735,000)	**(1,515,000)**
4	783,000	1,350,000	(567,000)	**(2,082,000)**
5	1,131,000	990,000	141,000	(1,941,000)
6	1,305,000	630,000	675,000	(1,266,000)
7	957,000	—	957,000	(309,000)
8	609,000	—	609,000	300,000

```
Peak funding requirement = (2,082,000), at the end of Month 4
Total cash in  = 480,000 + 5,220,000 = 5,700,000
Total cash out = 5,400,000
Closing position at Month 8 = 300,000
```

The closing 300,000 is not the margin. The other 300,000 is sitting in retention, released in two tranches — half at completion and half after the defects period — at which point the cumulative position reaches `300,000 + 150,000 + 150,000 = 600,000`, the margin, in cash, roughly a year after the work finished.

10.3 Profitable, and needing 2.08 million

The package makes 600,000 on 6,000,000 of value. To make it, the business must fund 2,082,000 at Month 4:

```
Peak funding ÷ margin = 2,082,000 ÷ 600,000 = 3.47
```

The funding requirement is about three and a half times the entire profit on the job. Nothing has gone wrong; this is what a normally performing contract looks like.

10.4 Sensitivity 1 — payment one month later

Everything unchanged except that receipts arrive three months after the work month rather than two:

Month	Cash in	Cash out	Cumulative cash
0	480,000	—	480,000
1	—	450,000	30,000
2	—	810,000	(780,000)
3	—	1,170,000	(1,950,000)
4	435,000	1,350,000	(2,865,000)
5	783,000	990,000	**(3,072,000)**
6	1,131,000	630,000	(2,571,000)
7	1,305,000	—	(1,266,000)
8	957,000	—	(309,000)
9	609,000	—	300,000

```
New peak funding requirement = (3,072,000), at the end of Month 5
Increase = 3,072,000 − 2,082,000 = 990,000
As a proportion = 990,000 ÷ 2,082,000 = 47.6 %
```

One month of payment timing raises the funding requirement by nearly half, and moves the trough a month later. Scope, programme, cost and margin are all identical. This is the number that should be in front of whoever negotiates payment terms.

10.5 Sensitivity 2 — no mobilisation advance

Remove the advance and its recovery. Net certificates rise to gross less retention only — 475,000, 855,000, 1,235,000, 1,425,000, 1,045,000 and 665,000 — still received two months in arrears:

Month	Cash in	Cash out	Cumulative cash
0	—	—	0
1	—	450,000	(450,000)
2	—	810,000	(1,260,000)
3	475,000	1,170,000	(1,955,000)
4	855,000	1,350,000	**(2,450,000)**
5	1,235,000	990,000	(2,205,000)
6	1,425,000	630,000	(1,410,000)
7	1,045,000	—	(365,000)
8	665,000	—	300,000

```
Peak without the advance = (2,450,000)
Benefit of the advance   = 2,450,000 − 2,082,000 = 368,000
```

The advance was 480,000, but it improves the funding position by only 368,000, because by the trough at Month 4 two certificates have already repaid part of it:

```
Recovery deducted from the Month 1 and Month 2 certificates = 40,000 + 72,000 = 112,000
480,000 − 112,000 = 368,000 ✓
```

An advance is worth less than its face value to the funding position, and how much less depends on the recovery rate and on where the trough falls. A forecast that credits the full advance overstates the funding benefit by 112,000 — which is `112,000 ÷ 480,000 = 23.3 %` of the advance.

Assumptions this example depends on. Full certification of applied value; a fixed two-month (or, in §10.4, three-month) payment lag; all outflows paid in the month incurred; no variations, escalation or indirect tax; retention released half at completion and half after the defects period, both outside the eight-month window shown; cost at a constant 90 % of value in every month. Relaxing any one of these moves the trough, which is exactly why sensitivities are run one variable at a time.

11. Checklist

Structure

- [ ] The cash forecast is a separate model from the cost forecast, driven by the same schedule.
- [ ] Gross value is phased from the time-phased baseline and the current progress forecast, not spread evenly.
- [ ] Retention, advance recovery and previous payments are modelled as separate lines, not netted.
- [ ] The payment date rule is stated explicitly, in periods, and matches the contract.
- [ ] Outflow terms are modelled by category — labour, subcontract, materials, plant — not as a single lag.

Reconciliations, every month

- [ ] Retention deducted to date equals the retention percentage times gross certified to date.
- [ ] Advance recovered to date is on track to repay the advance by completion.
- [ ] Total cash in over the life equals contract value, once retention is released.
- [ ] The closing cash position equals the margin, once all deductions have unwound.

Reporting

- [ ] The peak funding requirement is stated with its date.
- [ ] Sensitivities are run one variable at a time, and reported as depth and date of the trough.
- [ ] The payment-terms sensitivity is on the page every month, not on request.
- [ ] Retention outstanding is reported as a receivable with release dates and a named owner.
- [ ] Instructed-but-unpriced work is shown as cash exposure, not omitted because it has no certificate.

Judgement

- [ ] The certification assumption is stated, and compared against actual certification history.

- [ ] The forecast has been tested against the aggregate position of the portfolio, not just this project.
- [ ] Where the trough exceeds the available facility, the date it does so is escalated before it arrives.

Related

- `BPG-07 — Accruals and cut-off discipline` — the difference between cost incurred, cost invoiced and cost paid, which this guide depends on.
- `BPG-11 — Change orders and variations` — why an unpriced variation book is a funding problem as well as a commercial one.
- `BPG-09 — Estimate at completion: choosing and defending a method` — the cost forecast this model takes as its input.
- `BPG-14 — Monthly reporting that gets read` — how the peak funding requirement reaches a decision-maker in time to matter.
- `TPL-09 — Cash flow forecast` — the instrument implementing §10's cascade.

Sources and standards

The valuation, certification, retention and advance-recovery mechanisms described here follow the general structure used across construction and engineering contracts, including the standard forms published by bodies such as FIDIC and the NEC family; no clause, wording or numbering is reproduced, and the specific terms of any contract prevail over this description. Indirect tax and withholding treatment is jurisdiction specific and is a matter for the finance function. Internal references are BoK Domain 3 (Budgeting & Forecasting), BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 11 (Business Process Cycles). All figures in §10 are illustrative and were computed for this document.

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